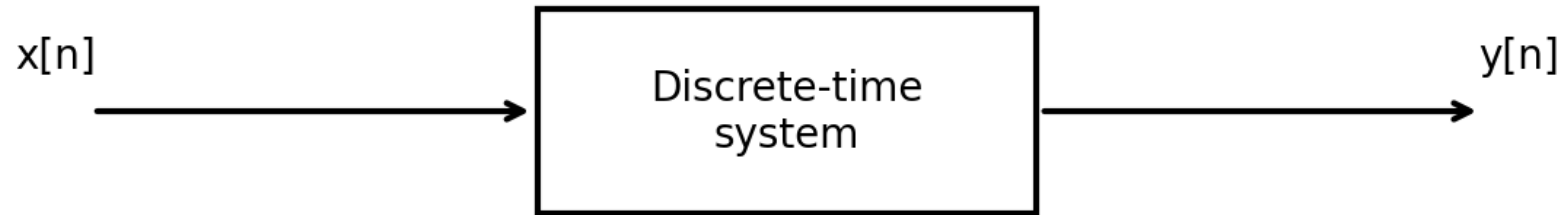


Topic 5

FIR Filters

Discrete-time systems, impulse response, convolution, and LTI systems



Key idea. An FIR filter computes each output sample from a finite set of present and past input samples.

Discrete-time systems

A system transforms an input sequence into an output sequence

$$y[n] = T\{x[n]\}$$

Example: $y[n] = a \cdot x[n]$

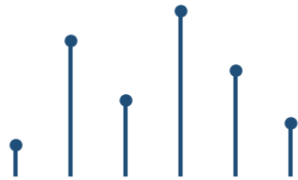
- A discrete-time system is a computational process acting on the input $x[n]$.
- The output $y[n]$ can scale, smooth, delay, combine, or otherwise transform the sequence.
- We study systems because filters are the building blocks of digital signal processing.

Key idea. A filter is best understood as an input-output rule.

Discrete-time systems — block diagram

Basic input-output view of a discrete-time system

input sequence $x[n]$



Discrete-time
system

$$y[n] = T\{x[n]\}$$

output sequence $y[n]$



A system transforms the input sequence $x[n]$ into the output sequence $y[n]$.

Moving-average filter

A simple FIR filter that smooths rapid fluctuations

3-point causal average: $y[n] = \frac{1}{3}(x[n] + x[n-1] + x[n-2])$

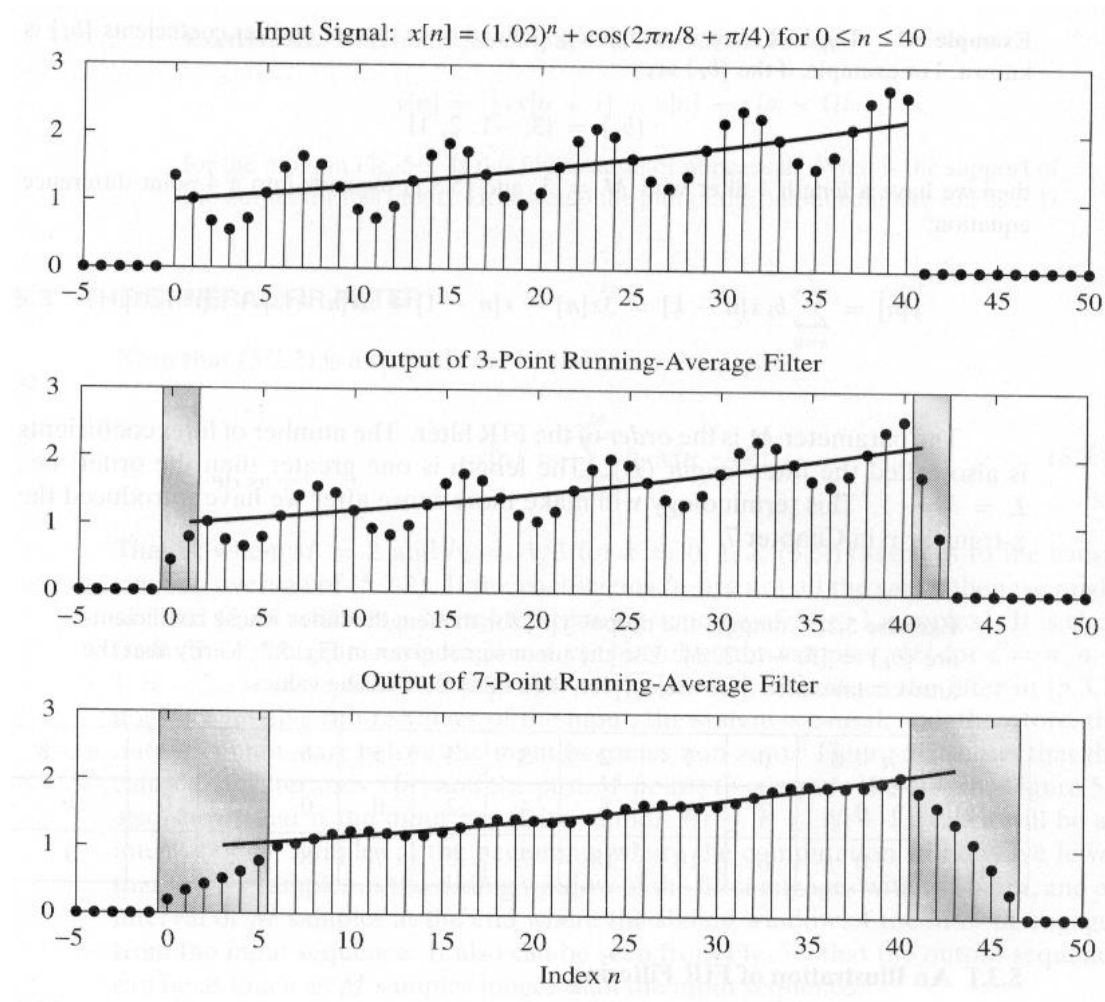
$$L - \text{point average: } y[n] = (1/L) \sum_{k=0}^{L-1} x[n-k]$$

- Each output sample is the average of a short window of neighboring input samples.
- Longer windows produce stronger smoothing but also more delay and less temporal detail.
- A centred moving average is usually noncausal because it needs future samples.

Key idea. A moving average smooths by spreading each sample across a short time window.

Moving-average filter — smoothing effect

Example of 3-point and 7-point running-average filtering



Longer moving-average windows produce stronger smoothing, but also more delay.

Example 1 — moving average

Computing the output of a causal 3-point smoother

Problem

Consider

$y[n] = \frac{1}{3}(x[n] + x[n-1] + x[n-2])$
and $x[n] = \{1, 2, 3, 2, 1\}$, starting at $n = 0$.

Compute $y[n]$ for all n where it can be nonzero.

Solution

- 1 Use the rule directly, sample by sample. Since the input is finite, take $x[n]=0$ outside $0 \leq n \leq 4$.
- 2 Start at $n=0$ and $n=1$:
 $y[0] = \frac{1}{3}(1+0+0) = \frac{1}{3}$, $y[1] = \frac{1}{3}(2+1+0) = 1$.
- 3 Middle samples, when the 3-point window is full:
 $y[2] = \frac{1}{3}(3+2+1) = 2$, $y[3] = \frac{1}{3}(2+3+2) = \frac{7}{3}$, $y[4] = \frac{1}{3}(1+2+3) = 2$.
- 4 Tail samples, after the input has ended:
 $y[5] = \frac{1}{3}(0+1+2) = 1$, $y[6] = \frac{1}{3}(0+0+1) = \frac{1}{3}$.

Final answer: $y[n] = \{\frac{1}{3}, 1, 2, \frac{7}{3}, 2, 1, \frac{1}{3}\}$. Each output is the average of the current and two previous input samples.

General FIR filter

Difference equation, coefficient list, and causality

$$y[n] = \sum_{k=0}^M b[k] \cdot x[n-k]$$

Order: M Length: $L = M + 1$

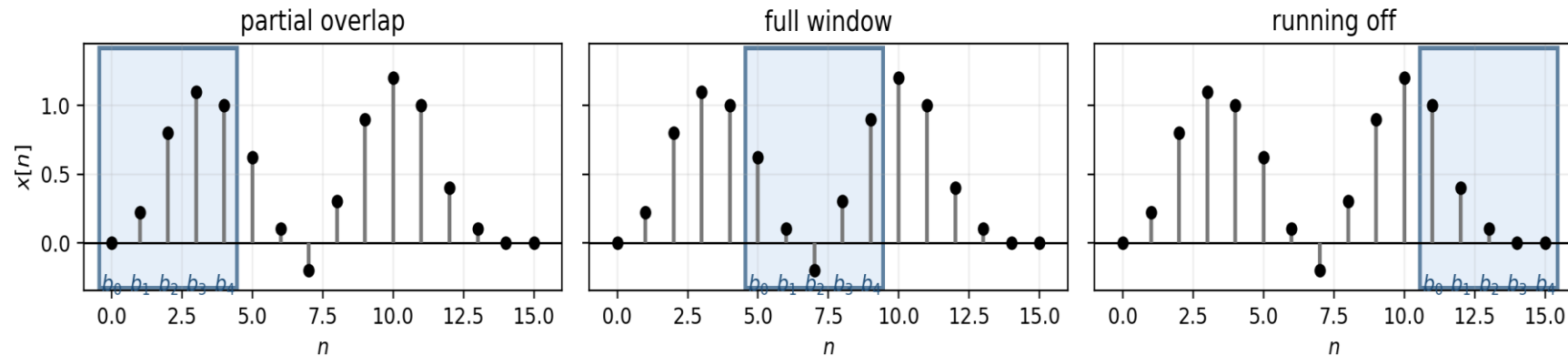
- The coefficients $b[0], \dots, b[M]$ specify how the delayed input samples are weighted.
- The filter is causal because it uses only the present sample and past samples $x[n-k]$.
- The coefficient list, difference equation, and impulse response are equivalent descriptions.

Key idea. Any causal FIR filter is a weighted sum of delayed copies of the input.

General FIR filter — window interpretation

The FIR output is a weighted sum over a finite window

Sliding FIR processing window



A causal FIR filter combines $M+1$ consecutive samples as the processing window slides along the data.

Example 2 — FIR coefficients and order

Reading the difference equation directly from the coefficient list

Problem

Let the FIR coefficients be $b[0]=3$, $b[1]=-1$, $b[2]=2$, and $b[3]=1$. Write the difference equation and identify the order and length.

Solution

- 1 Insert the coefficients into $y[n] = \sum b[k]x[n-k]$.
- 2 This gives $y[n] = 3x[n] - x[n-1] + 2x[n-2] + x[n-3]$.
- 3 There are four coefficients, so the filter length is $L = 4$.
- 4 The highest delay is 3, so the filter order is $M = 3$.

Final answer: $y[n] = 3x[n] - x[n-1] + 2x[n-2] + x[n-3]$, with order $M = 3$ and length $L = 4$.

Key idea. For FIR filters, the order is the largest delay and the length is the number of taps.

Unit impulse and impulse response

The building blocks of FIR and LTI analysis

$\delta[n] = 1$ at $n=0$, and 0 otherwise

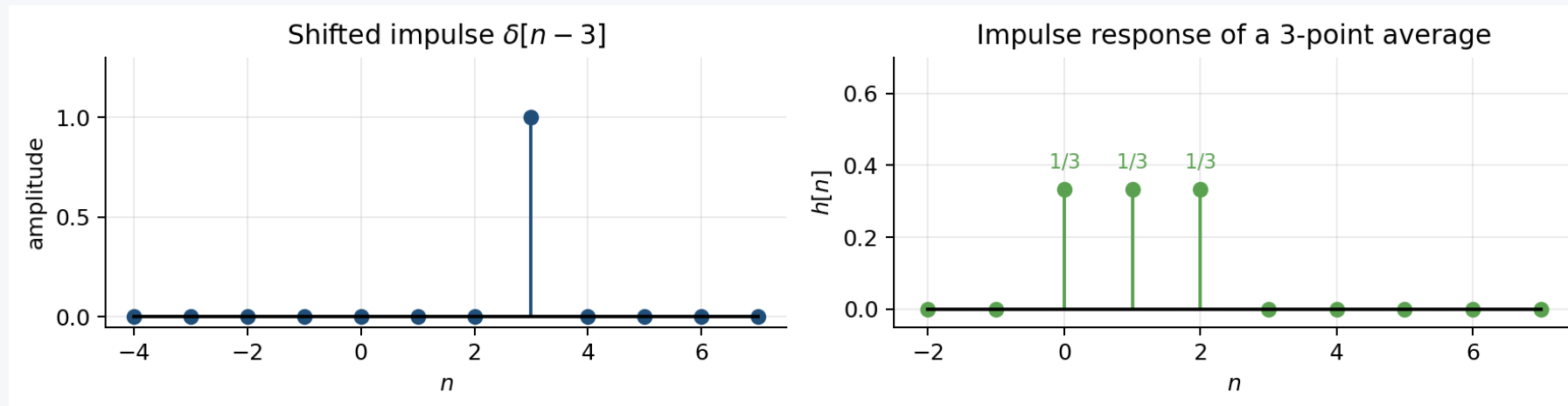
$h[n]$ = output when $x[n] = \delta[n]$

- A shifted impulse $\delta[n-n_0]$ is nonzero only at $n = n_0$.
- Any sequence can be written as $x[n] = \sum x[k]\delta[n-k]$, a sum of scaled, shifted impulses.
- For a causal FIR filter, $h[n]$ is exactly the coefficient sequence placed at the appropriate delays.
- A unit delay is the simplest FIR system: $y[n] = x[n-1]$.

Key idea. Knowing the impulse response means knowing the filter.

Unit impulse and impulse response — visual examples

A shifted impulse and the impulse response of a 3-point average



The unit impulse is the building block of signals, and the impulse response fully characterizes an LTI FIR filter.

Example 3 — impulse response

From the difference equation to $h[n]$ and shifted-impulse outputs

Problem

A causal FIR system is $y[n] = 2x[n] - x[n-1] + (1/2)x[n-3]$. Find $h[n]$, and then the output for input $\delta[n-4]$.

Solution

- 1 Set $x[n] = \delta[n]$; the impulse response is $h[n] = 2\delta[n] - \delta[n-1] + (1/2)\delta[n-3]$.
- 2 Thus the nonzero samples are $h[0]=2$, $h[1]=-1$, and $h[3]=1/2$.
- 3 A delayed impulse at the input produces the same delay at the output.
- 4 Therefore the response to $\delta[n-4]$ is $h[n-4] = 2\delta[n-4] - \delta[n-5] + (1/2)\delta[n-7]$.

Final answer: $h[n] = 2\delta[n] - \delta[n-1] + (1/2)\delta[n-3]$, and for input $\delta[n-4]$ the output is $2\delta[n-4] - \delta[n-5] + (1/2)\delta[n-7]$.

Convolution and FIR filters

A compact way to compute the output of a filter

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n - k]$$

For finite sequences: $\text{length}(y) = \text{length}(x) + \text{length}(h) - 1$

- Convolution adds scaled and shifted copies of one sequence, weighted by the other.
- For FIR filters, convolution and the difference equation produce exactly the same output.
- Convolution is commutative: $x * h = h * x$. It is also associative.

Key idea. Convolution is the natural algebra of LTI systems and FIR filtering.

Example 4 — numerical convolution

Computing the FIR output sample by sample

Problem

Let $x[n] = \{2, 4, 6, 4, 2\}$ and $h[n] = \{3, -1, 2, 1\}$, both starting at $n = 0$. Compute $y[n] = x[n] * h[n]$.

Solution

- 1 The output length is $5 + 4 - 1 = 8$ samples.
- 2 Start at the edges: $y[0] = 2 \cdot 3 = 6$ and $y[1] = 4 \cdot 3 + 2 \cdot (-1) = 10$.
- 3 Continue the overlap-add calculation or use the shifted-copy view of convolution.
- 4 The full result is $y[n] = \{6, 10, 18, 16, 18, 12, 8, 2\}$.

Final answer: $y[n] = \{6, 10, 18, 16, 18, 12, 8, 2\}$. Reversing the order to $h * x$ gives the same result.

Convolution — numerical interpretation

Summing scaled and shifted copies of the input

Each row is one shifted, scaled copy of the input; the last row is their sum.

n	0	1	2	3	4	5	6	7
$h[0]x[n-0]$	6	12	18	12	6	0	0	0
$h[1]x[n-1]$	0	-2	-4	-6	-4	-2	0	0
$h[2]x[n-2]$	0	0	4	8	12	8	4	0
$h[3]x[n-3]$	0	0	0	2	4	6	4	2
$y[n]$	6	10	18	16	18	12	8	2

This table shows how each shifted term contributes to the final convolution output $y[n]$.

FIR-filter implementation

Direct-form realization uses delays, multipliers, and adders

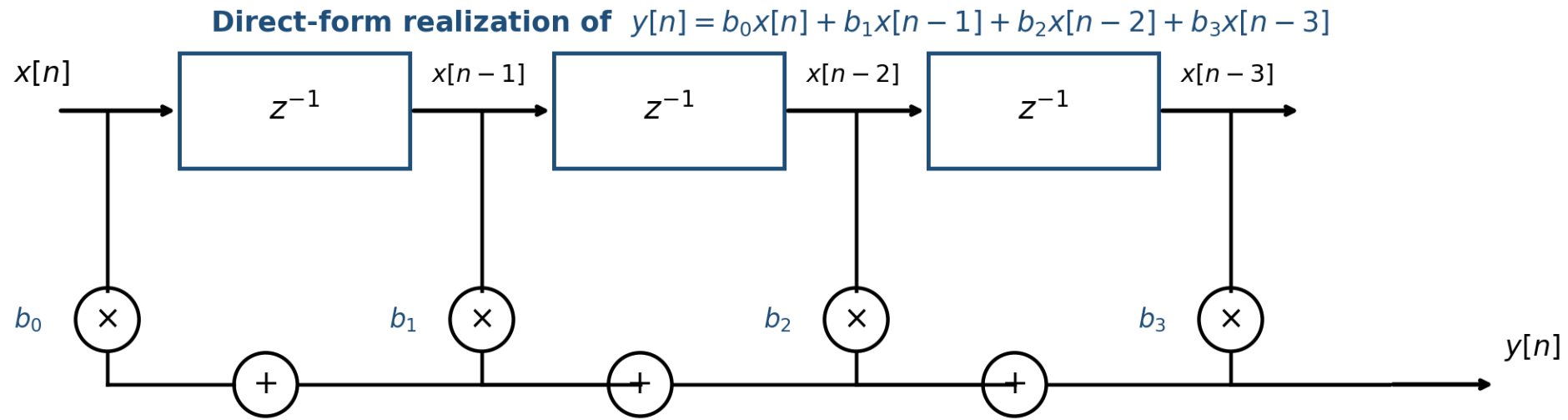
Direct form implements $y[n] = b[0]x[n] + b[1]x[n-1] + \dots + b[M]x[n-M]$

- The input passes through a chain of unit delays to create $x[n-1]$, $x[n-2]$, ..., $x[n-M]$.
- Each branch is multiplied by its coefficient $b[k]$, and the branches are summed.
- A length- L FIR filter needs $M = L-1$ delay elements, L multipliers, and $L-1$ additions.
- Alternative block diagrams are equivalent if they implement the same input-output rule.

Key idea. Implementation details may change, but the realized FIR system stays the same.

FIR-filter implementation — direct form

Realization with unit delays, multipliers, and adders



A direct-form FIR implementation creates delayed versions of $x[n]$, weights them by $b[k]$, and sums the branches.

Linearity and time invariance

Why FIR systems are easy to analyze

Linearity: $T\{\alpha x_1[n] + \beta x_2[n]\} = \alpha T\{x_1[n]\} + \beta T\{x_2[n]\}$

Time invariance: if $y[n] = T\{x[n]\}$, then $T\{x[n - n_0]\} = y[n - n_0]$

- Linearity combines homogeneity and additivity: scale inputs and outputs scale the same way.
- Time invariance means a time shift at the input produces the same shift at the output.
- The general FIR system is LTI because its delays and coefficients are fixed.

Key idea. LTI properties make convolution possible and give a complete system description through $h[n]$.

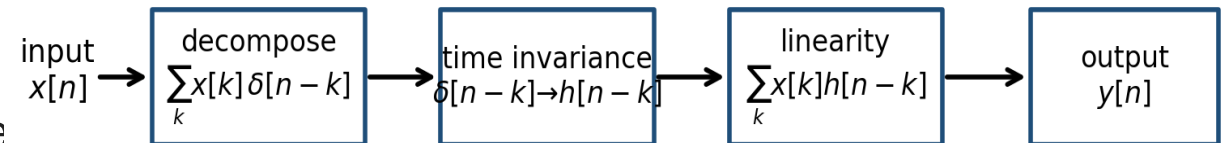
Convolution and LTI systems

The impulse response completely characterizes an LTI system

$$x[n] = \sum_k x[k] \cdot \delta[n - k]$$
$$y[n] = \sum_k x[k] \cdot h[n - k]$$

- Decompose the input into scaled, shifted impulses.
- By time invariance, $\delta[n-k]$ produces $h[n-k]$.
- By linearity, the total output is the sum of those responses.
- Therefore every LTI system is represented by a convolution with $h[n]$.

LTI viewpoint behind convolution



Every scaled, shifted impulse produces a scaled, shifted copy of the impulse response.

Key idea. Impulse decomposition + LTI properties = convolution sum.

Cascaded LTI systems

Two FIR filters in sequence are equivalent to a single FIR filter

$$h_{eq}[n] = h_1[n] * h_2[n]$$

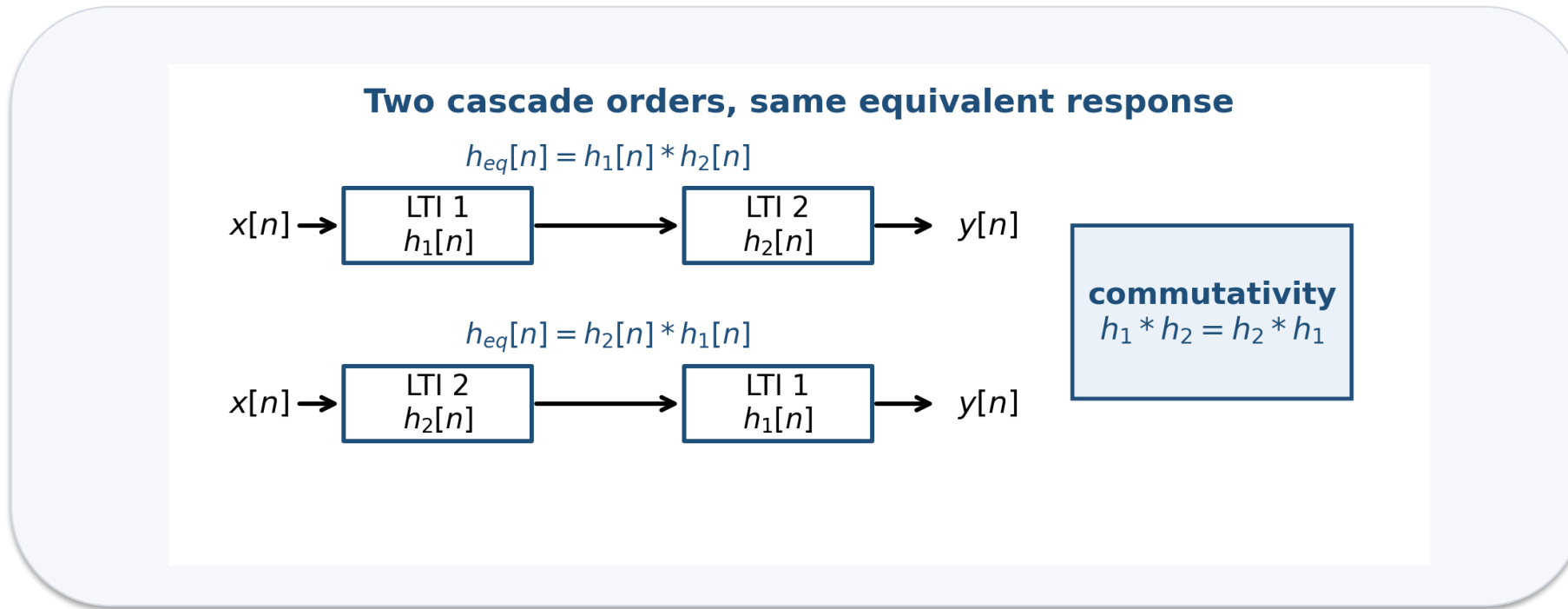
Because convolution is commutative: $h_1 * h_2 = h_2 * h_1$

- Placing one LTI system after another produces a new LTI system.
- The equivalent impulse response is the convolution of the individual impulse responses.
- The order of the cascade does not change the overall result.

Key idea. Cascades let us build complicated filters from simpler ones.

Cascaded LTI systems — equivalent response

Why the order of two cascaded FIR systems does not matter



Convolution is commutative, so the cascade order can be interchanged.

Example 5 — cascaded FIR systems

Finding the equivalent impulse response of a cascade

Problem

Two causal FIR systems are connected in cascade with $h_1[n] = \{1, 1, 1, 1\}$ and $h_2[n] = \{1, 1, 1\}$, both starting at $n = 0$. Find $h_{eq}[n]$.

Solution

- 1 Convolve the two moving-sum responses: $h_{eq}[n] = h_1[n] * h_2[n]$.
- 2 The overlap grows as 1, 2, 3, then stays at 3, and finally decreases as 2, 1.
- 3 So $h_{eq}[n] = \{1, 2, 3, 3, 2, 1\}$.
- 4 The length is $4 + 3 - 1 = 6$, so the equivalent order is $M = 5$.

Final answer: $h_{eq}[n] = \{1, 2, 3, 3, 2, 1\}$. The same result is obtained if the two systems are interchanged.

Key idea. Commutativity of convolution explains why the order of two cascaded LTI systems does not matter.

Notation and takeaways

Minimum language for FIR filters and convolution

$x[n]$	input sequence
$y[n]$	output sequence
$b[k]$	FIR coefficients
M	filter order
L	filter length = $M+1$

$\delta[n]$	unit impulse
$h[n]$	impulse response
$*$	convolution
LTI	linear and time invariant
causal	uses only present/past inputs